

Администрирование сетевых подсистем

Лабораторная работа №8 — Настройка SMTP-сервера Postfix

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Цели и задачи работы

Приобретение практических навыков по установке и конфигурированию SMTP-сервера Postfix.

Выполнение лабораторной работы

Установка и первичная настройка Postfix

```
Total 7.3 MB/s | 2.2 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : postfix-lmdb-2:3.8.5-8.el10.x86_64 1/3
  Running scriptlet: postfix-2:3.8.5-8.el10.x86_64 2/3
  Installing     : postfix-2:3.8.5-8.el10.x86_64 2/3
  Running scriptlet: postfix-2:3.8.5-8.el10.x86_64 2/3
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64 3/3
  Installing     : s-nail-14.9.24-12.el10.x86_64 3/3
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64 3/3

Installed:
postfix-2:3.8.5-8.el10.x86_64      postfix-lmdb-2:3.8.5-8.el10.x86_64      s-nail-14.9.24-12.el10.x86_64

Complete!
[root@server.manovichkov.net ~]# firewall-cmd --add-service=smtp
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns http https smtp ssh
[root@server.manovichkov.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfine
d_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.manovichkov.net ~]# systemctl enable postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/postfix.service'.
```

Рис. 1: Установка и запуск Postfix

Изменение параметров Postfix

```
unverified_sender_defer_code = 450
unverified_sender_reject_code = 450
unverified_sender_reject_reason =
unverified_sender_tempfail_action = $reject_tempfail_action
use_srv_lookup =
verp_delimiter_filter = -+=
virtual_alias_address_length_limit = 1000
virtual_alias_domains = $virtual_alias_maps
virtual_alias_expansion_limit = 1000
virtual_alias_maps = $virtual_maps
virtual_alias_recursion_limit = 1000
virtual_delivery_slot_cost = $default_delivery_slot_cost
virtual_delivery_slot_discount = $default_delivery_slot_discount
virtual_delivery_slot_loan = $default_delivery_slot_loan
virtual_delivery_status_filter = $default_delivery_status_filter
virtual_destination_concurrency_failed_cohort_limit = $default_destination_concurrency_failed_cohort_limit
virtual_destination_concurrency_limit = $default_destination_concurrency_limit
virtual_destination_concurrency_negative_feedback = $default_destination_concurrency_negative_feedback
virtual_destination_concurrency_positive_feedback = $default_destination_concurrency_positive_feedback
virtual_destination_rate_delay = $default_destination_rate_delay
virtual_destination_recipient_limit = $default_destination_recipient_limit
virtual_extra_recipient_limit = $default_extra_recipient_limit
virtual_gid_maps =
virtual_initial_destination_concurrency = $initial_destination_concurrency
virtual_mailbox_base =
virtual_mailbox_domains = $virtual_mailbox_maps
virtual_mailbox_limit = 51200000
virtual_mailbox_lock = fcntl, dotlock
virtual_mailbox_maps =
virtual_minimum_delivery_slots = $default_minimum_delivery_slots
virtual_minimum_uid = 100
virtual_recipient_limit = $default_recipient_limit
virtual_recipient_refill_delay = $default_recipient_refill_delay
virtual_recipient_refill_limit = $default_recipient_refill_limit
virtual_transport = virtual
virtual_transport_rate_delay = $default_transport_rate_delay
virtual_uid_maps =
[root@server.manovichkov.net ~]#
```

```
[root@server.manovichkov.net ~]# postconf mydomain
mydomain = manovichkov.net
[root@server.manovichkov.net ~]# postconf -e 'myorigin = $mydomain'
[root@server.manovichkov.net ~]# postconf mydomain
mydomain = manovichkov.net
[root@server.manovichkov.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.manovichkov.net ~]# postfix check
[root@server.manovichkov.net ~]# systemctl reload postfix.service
[root@server.manovichkov.net ~]#
```

Рис. 3: Настройка протоколов IPv4 для Postfix

Проверка работы Postfix

```
[manovichkov@server.manovichkov.net ~]$ sudo tail -f /var/log/maillog
[sudo] password for manovichkov:
Oct  8 10:06:35 server postfix/postfix-script[12088]: refreshing the Postfix mail system
Oct  8 10:06:35 server postfix/master[11117]: reload -- version 3.8.5, configuration /etc/postfix
Oct  8 10:06:35 server postfix/master[11117]: warning: ignoring inet_protocols parameter value change
Oct  8 10:06:35 server postfix/master[11117]: warning: old value: "all", new value: "ipv4"
Oct  8 10:06:35 server postfix/master[11117]: warning: to change inet_protocols, stop and start Postfix
Oct  8 10:07:12 server postfix/pickup[12093]: CE667609E765: uid=1001 from=<manovichkov>
Oct  8 10:07:12 server postfix/cleanup[12197]: CE667609E765: message-id=<20251008100712.CE667609E765@server.manovichkov.net>
Oct  8 10:07:12 server postfix/qmgr[12094]: CE667609E765: from=<manovichkov@manovichkov.net>, size=352, nrcpt=1 (queue active)
Oct  8 10:07:12 server postfix/local[12200]: CE667609E765: to=<manovichkov@server.manovichkov.net>, relay=local, delay=0.01, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct  8 10:07:12 server postfix/qmarf120941: CE667609E765: removed
```

Рис. 4: Мониторинг доставки письма на сервере


```
[root@server.manovichkov.net ~]# postconf inet_interfaces
inet_interfaces = localhost
[root@server.manovichkov.net ~]# postconf -e 'inet_interfaces = all'
[root@server.manovichkov.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.manovichkov.net ~]# postfix check
[root@server.manovichkov.net ~]# systemctl reload postfix.service
[root@server.manovichkov.net ~]# systemctl stop postfix.service
■
```

Рис. 5: Лог успешной доставки письма с клиента

Настройка сетевых параметров для пересылки

```
Oct  8 10:13:22 server postfix/smtpd[13226]: connect from client.manovichkov.net[192.168.1.30]
Oct  8 10:13:22 server postfix/smtpd[13226]: DCE6F609E75D: client=client.manovichkov.net[192.168.1.30]
Oct  8 10:13:22 server postfix/cleanup[13230]: DCE6F609E75D: message-id=<20251008101322.D2D5B5CDD@client.manovichkov.net>
Oct  8 10:13:22 server postfix/smtpd[13226]: disconnect from client.manovichkov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Oct  8 10:13:22 server postfix/qmgr[13213]: DCE6F609E75D: from=<manovichkov@client.manovichkov.net>, size=580, nrcpt=1 (queue active)
Oct  8 10:13:22 server postfix/local[13231]: DCE6F609E75D: to=<manovichkov@server.manovichkov.net>, relay=local, delay=0.01, delays=0/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct  8 10:13:22 server postfix/qmgr[13213]: DCE6F609E75D: removed
```

Рис. 6: Настройка параметров mynetworks и inet_interfaces

Настройка работы с доменом manovichkov.net

```
Oct  8 10:13:22 server postfix/smtpd[13226]: disconnect from client.manovichkov.net[192.168.1.30] ehlo=2 starttls=1 mail
=1 rcpt=1 data=1 quit=1 commands=7
Oct  8 10:13:22 server postfix/qmgr[13213]: DCE6F609E75D: from=<manovichkov@client.manovichkov.net>, size=580, nrcpt=1 (
queue active)
Oct  8 10:13:22 server postfix/local[13231]: DCE6F609E75D: to=<manovichkov@server.manovichkov.net>, relay=local, delay=0
.01, delays=0/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct  8 10:13:22 server postfix/qmgr[13213]: DCE6F609E75D: removed
```

Рис. 7: Ошибка доставки до настройки MX-записи

Добавление MX-записей в DNS

```
manovichkov.net [----] 0 L:[ 1+23 24/ 24] *(601 / 601b) <EOF>
$ORIGIN .
$TTL 86400<---->; 1 day
manovichkov.net<----->IN SOA<----->manovichkov.net. server.manovichkov.net. (
<-----><-----><-----><----->2025091713 ; serial
<-----><-----><-----><----->86400 ; refresh (1 day)
<-----><-----><-----><----->3600 ; retry (1 hour)
<-----><-----><-----><----->604800 ; expire (1 week)
<-----><-----><-----><----->10800 ; minimum (3 hours)
<-----><-----><-----><----->)
<-----><-----><----->NS<----->manovichkov.net.
<-----><-----><----->A<----->192.168.1.1
<-----><-----><----->MX 10<->mail.manovichkov.net.
$ORIGIN manovichkov.net.
$TTL 1200<---->; 20 minutes
client<-><-----><----->A<----->192.168.1.30
<-----><-----><-----><----->DHCID<->( AAEB1FhaDarqBrLi004j6+E00+13CnZ6kVJkXXSH25vj
<-----><-----><-----><----->a6Y= ) ; 1 1 32
$TTL 86400<---->; 1 day
dhcp<-><-----><----->A<----->192.168.1.1
ns<-><-----><----->A<----->192.168.1.1
server<-><-----><----->A<----->192.168.1.1
www<-><-----><----->A<----->192.168.1.1
mail<-><-----><----->A<----->192.168.1.1
```

Рис. 8: Файл прямой зоны

Добавление MX-записей в DNS

```
root@server:/var/named/master/rz - sudo -i
root@server:/var/named/master/rz - sudo -i  manovichkov@server:~  manovichkov@server:~ - sudo tail -f /var/log/
192.168.1.1 [~M~] 50 L:[ 1+18 19/ 24] *(554 / 687b) 0010 0x00A [*][X]
$ORIGIN .
$TTL 86400<----->; 1 day
1.168.192.in-addr.arpa<IN SOA>1.168.192.in-addr.arpa. server.manovichkov.net. (
<-----><-----><-----><----->2025091709 ; serial
<-----><-----><-----><----->86400 ; refresh (1 day)
<-----><-----><-----><----->3600 ; retry (1 hour)
<-----><-----><-----><----->604800 ; expire (1 week)
<-----><-----><-----><----->10800 ; minimum (3 hours)
<-----><-----><-----><----->)
<-----><-----><-----><----->NS<----->1.168.192.in-addr.arpa.
<-----><-----><-----><----->A<----->192.168.1.1
<-----><-----><-----><----->PTR<----->server.manovichkov.net.
<-----><-----><-----><----->MX 10<----->mail.manovichkov.net.
$ORIGIN 1.168.192.in-addr.arpa.
1<-----><-----><-----><----->PTR<----->server.manovichkov.net.
<-----><-----><-----><----->PTR<----->ns.manovichkov.net.
<-----><-----><-----><----->PTR<----->dhcp.manovichkov.net.
<-----><-----><-----><----->PTR<----->www.manovichkov.net.
<-----><-----><-----><----->PTR<----->mail.manovichkov.net.
$TTL 1200<----->; 20 minutes
30<-----><-----><-----><----->PTR<----->client.manovichkov.net.
<-----><-----><-----><----->DHCID<----->( AAEBLfhaDqrBrLi004j6+E00+13CnZ6kVJkXXSH25vj
<-----><-----><-----><----->a6Y= ) ; 1 1 32
```

Рис. 9: Файл обратной зоны

Корректировка конфигурации Postfix

```
[root@server.manovichkov.net rz]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
[root@server.manovichkov.net rz]# postfix check
[root@server.manovichkov.net rz]# systemctl reload postfix.service
[root@server.manovichkov.net rz]# restorecon -vR /etc
[root@server.manovichkov.net rz]# restorecon -vR /named
restorecon: lstat(/named) failed: No such file or directory
[root@server.manovichkov.net rz]# restorecon -vR /var/named/
[root@server.manovichkov.net rz]# systemctl restart named
[root@server.manovichkov.net rz]#
```

Рис. 10: Перезагрузка служб и проверка конфигурации

Успешная доставка писем после настройки

```
Oct  8 10:18:42 server postfix/smtpd[14085]: disconnect from client.manovichkov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Oct  8 10:18:42 server postfix/qmgr[13989]: B3B3A609E760: from=<manovichkov@client.manovichkov.net>, size=580, nrcpt=1 (queue active)
Oct  8 10:18:42 server postfix/local[14089]: B366C609E75F: to=<manovichkov@server.manovichkov.net>, relay=local, delay=0.01, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct  8 10:18:42 server postfix/qmgr[13989]: B366C609E75F: removed
Oct  8 10:18:42 server postfix/local[14089]: B3B3A609E760: to=<manovichkov@server.manovichkov.net>, relay=local, delay=0.01, delays=0/0.01/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct  8 10:18:42 server postfix/qmgr[13989]: B3B3A609E760: removed
```

Рис. 11: Успешная передача письма с клиента

Выводы

- Настроен и протестирован SMTP-сервер Postfix
- Реализована работа почты в локальной сети
- Добавлены MX-записи для домена `manovichkov.net`
- Проверена доставка писем между узлами
- Создан скрипт автоматической установки и настройки почтового сервера